sEMG-Based Gesture Recognition Using Deep Learning From Noisy Labels

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sEMG-Based Gesture Recognition Using Deep Learning From Noisy Labels

Akram Fatayer, Wenpeng Gao, and Yili Fu

***Abstract*—Gesture recognition for myoelectric prosthesis control utilizing sparse multichannel surface Electromyography (sEMG) is a challenging task, and from a Muscle-Computer Interface (MCI) standpoint, the performance is still far from optimal. However, the design of a well-performed sEMG recognition system depends on the flexibility of the input-output function and the dataset's quality. To improve the performance of MCI, we proposed a novel gesture recognition framework that (i) Enrich the spectral information of the sparse sEMG signals by constructing a fused map image (denoted as sEMG-Map) that integrates a multiresolution decomposition (by means of orthogonal wavelets) through the raw signals then rely upon the Convolutional Neural Network (CNN) capacity to exploit the composite hierarchies in the constructed sEMG-Map input. (ii) Deals with the label noise by proposing a data-centric method (denoted as ALR-CNN) that synchronously refines the falsely labeled samples and optimizes the CNN model based on two basic assumptions. First, the deep model accuracy improves as the training progress. Second, a set of successive learnable max-activated outputs of a well-performed deep model is a reliable estimator for motion detection in the muscle activation pattern. Our proposed framework is evaluated on three large-scale public databases. The average classification accuracy is 95.50%, 95.85%, and 85.58% for NinaPro DB2, NinaPro DB7, and NinaPro DB3, respectively. The experimental results verify the effectuality of the proposed method and show high accuracy.**

***Index Terms*—Deep learning, gesture recognition, label noise, motion detection, sEMG.**

I. Introduction

Electromyography (EMG) is the electrical action potential attributed to the Motor Unit (MU), which is the nerve system's final common path and the neuromuscular system's basic functional unit. MU is comprised of the entire motoneuron and the innervated muscle fibers by the motoneuron's axon. The muscle fibers innervated by a single axon are known as the muscle unit. The cumulative contributions of the activated muscle unit discharge generate a motor unit action potential (MUAP). Therefore, surface EMG (sEMG) can be seen as a window to the underlying mechanism of the nerve system operation that allows extracting information about the movement control [1].

Representing the sEMG signals into inherent functionality that conveys the user's movement intention is the aim of the Muscle-Computer interface (MCI) that uses sEMG-based gesture recognition. However, designing a well-performed sEMG recognition system depends on:

1) The flexibility of the input-output function (i.e., the extracted features, the classification algorithm, or both.)
2) The quality of the data.

Problems like sEMG-based gesture recognition necessitate a flexible input-output function, which can resist irrelevant input variations. These include but are not limited to a variety of subcutaneous thicknesses of the tissue, anatomy of muscles, or location of active motor units within the recorded muscle. At the same time, it is sensitive to minute variation caused by neurophysiological events.

However, conventional machine learning algorithms have limited capabilities to curve their input space and process biological data in their raw form [2] due to the data's complexity (e.g., the nonstationary, non-linear, and stochastic nature of the EMG signal [3], [4]). Therefore, a recognition system based on conventional machine learning usually requires a pre-processing step. Fig. 1 shows a common pipeline of gesture recognition. One drawback of such a representation of the signal is that some subtle yet beneficial information could be lost, especially when the problem complexity increases (e.g., the number of gestures is increased).

On the other hand, deep learning has the capacity to process natural signals in their raw form. Nevertheless, the performance of deep learning methods such as Convolutional Neural Network (CNN) for gesture recognition based on sparse multichannel sEMG, which is the most common recording technique used in wearable systems [5], is still far from optimal. That might be because the sparse multichannel sEMG can only capture limited information in comparison with High-Density sEMG (HD-sEMG), which employs hundreds of electrodes. Therefore, in this work, we try to enrich the sEMG representation by means

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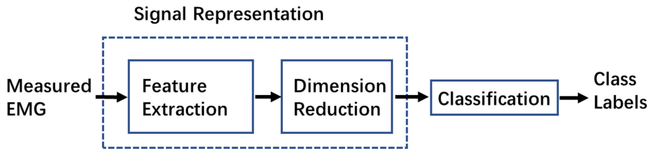

Fig. 1. Pipeline of a classical gesture recognition using sEMG.

of Wavelet Packet Transform (WPT). We transfer the problem of signal classification to that of image classification, in which the concept of a windowed sEMG image fusion through multiresolution decomposition (denoted as sEMG-Map) is introduced to generate a rich-detailed map for each processing window. Subsequently, the map is fed to a deep learning-based image classifier for gesture recognition.

Nevertheless, real datasets are inherently subject to corruption and noise, which may degrade the classifier's performance [6]. Many factors described in [7] were used to evaluate the data quality. The label noise is, by far, the most frequent type of noise, and it is the only concern of this part of the study. Therefore, we also proposed a novel Auto-Label-Refining CNN (ALR-CNN) method to deal with the label noise by refining the movement boundaries in an end-to-end learning structure.

In summary, the contributions of this paper could be concluded in the following points:

1) sEMG-Map is introduced, which is fused of a windowed raw sEMG and the coefficients of a multilevel wavelet decomposition. The map contains a comprehensive piece of information about the recorded MUAPs and the muscle synergy.
2) A novel data-centric method based on CNN model (denoted as ALR-CNN) is developed to handle the problem of label noise by synchronously refining the mislabeling samples and optimizing the model.
3) Comprehensive assessments on three sparse sEMG databases are performed to evaluate the proposed gesture recognition framework. The experimental results show that the proposed framework is comparable to or even outperforms the state-of-the-art deep learning methods. Moreover, it shows the learning capability of the proposed framework in the presence of noisy labels.

The remainder of this paper is organized as follows. Section II reviews the related works; Section III formulates the learning problem; Section IV overviews our proposed method; Section V introduces the input construction process; Section VI details the proposed ALR-CNN method; Section VII demonstrates the experimental results; Section VIII discusses the results, limitations and future work, and section IX concludes the paper.

II. Related Work

As previously stated, a well-performed sEMG recognition system relies on the appropriateness of the classification method and the quality of the data which the classifier trained upon it. In this section, we first briefly review the classification methods used for gesture recognition based on sEMG. After that, we review the state-of-the-art in handling label noise in machine learning and computer vision. In addition, we debriefed the motion detection methods in the sEMG signal since the label noise in the case of sEMG arises due to the challenge of accurately identifying the exact onset and offset times.

A. Classification Methods

The classification methods can be broadly divided into deep learning methods and conventional machine learning methods. Conventional machine learning methods assume that the raw sEMG is inappropriate for gesture recognition because of its complex nature. Therefore, the feature plays a critical role in the success of conventional machine learning methods. Human specialists have proposed a variety of sEMG feature sets based on feature types such as time-domain, frequency-domain, and time-frequency [8]–[15]. Conventional machine learning methods based on sparse multichannel sEMG can achieve acceptable performance for a relatively small number of movements (e.g., four to twelve movements) [11], [16], [17]. However, classic feature sets suffer when the problem's complexity increases (e.g., the number of movements to be classified or the performed movement's complexity increases). For instance, Atzori *et al.* [10] evaluated various features on their proposed NinaPro databases using a conventional machine learning framework and the best-obtained recognition result over the second database DB2 for 50 movements was only 75.27%.

Deep learning methods, such as CNNs, have lately been used in sEMG-based gesture recognition systems [18], [19]. In spite of the success of CNN methods for gesture recognition utilizing the raw HD-sEMG directly [18], [20], their performance is still far from optimal when sparse multichannel sEMG is used. For example, on the second sparse multichannel NinaPro database DB2, Atzori *et al.* [19] produced a baseline average gesture recognition accuracy of 60.3% by feeding the raw sEMG into CNN; Zhai *et al.* [21] achieved average recognition accuracy of 78.7% by feeding the sEMG's spectrogram into a CNN; while Wei *et al.* [22] evaluated 11 classical feature sets and reached an average gesture recognition accuracy of 83.7% by aggregating the views obtained from parallel multistream CNNs (each CNN fed by a performance-based selected feature) into a multi-view deep learning framework. However, it is worth mentioning that Wei *et al.* reported that the feature sets are highly relevant to the datasets. For example, one of the feature sets achieved an average accuracy of 83.7% over the NinaPro DB1 dataset, while the same feature set achieved only 8.1% over the NinaPro DB2 dataset.

B. Learning With Noisy Label

In neuroscience and biomedical recognition tasks, it is imperative to obtain large and precisely labeled datasets. Considerable efforts have been offered in machine learning and computer vision literature to deal with label noise. Some methods use sample weights obtained from the network's performance to decrease the effect of noisy labels [23], [24]. Other techniques use auxiliary networks to learn the pattern of the noisy labels in order to apply sample weights [25], [26]. Several methods suggest using altered loss functions to reduce the impact of noisy labels during training [27]–[29]. Brodley *et al.* eliminate potentially noisy labels from training [30]. Jan *et al.* use model uncertainty to identify and filter out label noise [31].

Moreover, the label noise in gesture recognition based on sEMG also can be viewed as a motion detection problem. In the literature of this context, some authors attempted to discard the transition processes between movements where the label noise exists [16], [32], with the drawback that information from the discarded portion of the movement is deserted, which could affect the realistic performance of the gesture recognition method. Others attempted to utilize automatic motion detection methods such as moving average [33], standard deviation [34], [35], and Generalized Likelihood Ratio (GLR) [36]. Concerning the NinaPro datasets, the previous works relied on the GLR algorithm utilized by Kuzborskij *et al.* in order to correct the erroneous movement labels [10], [19], [21], [22], [36].

III. Problem Statement

The problem of sEMG-based gesture recognition using deep learning can be formulated as follows:

$$m_i = h_\omega(x_i, l_i) \tag{1}$$

where x and l denote the CNN's input and the corresponding label, respectively, h_ω refers to the CNN model with the learned parameters ω, and m refers to the network's output softmax scores. Supervised learning is the common way to train the model to find the optimal parameters ω with labeled data.

In previous works, x is usually formed from the raw sEMG signals directly [18]–[20]. In contrast, Zhai *et al.* attempted to extract the sEMG spectrogram to form x [21], while Wei *et al.* extracted a set of classical sEMG features and parallelly fed each feature into multiple CNNs, then fused the learned features through a multi-view aggregation network [22].

In this work, instead of feeding the raw sEMG directly [18]–[20] or reducing the signal's complexity by replacing the sEMG signals with some dedicated feature sets [21], [22], we aimed to enhance the spectral information of the recorded sEMG signal by injecting a multiresolution representation utilizing multilevel wavelet decomposition through the windowed sEMG signal to construct a rich-detailed fused map and depends on the CNN model to elaborate the high redundant structures in the formed map-image. The input construction process is introduced in section V.

Moreover, in the noisy label learning scenario, the given label l_i may not be accurate, which shall seriously degenerate classifiers' performance. In sEMG-based gesture recognition tasks, the label noise emerges mainly due to human reaction time and experiment conditions, where subjects may not correctly match the performed movements with the stimuli guided by the experiments' software. Therefore, in this work, we also implemented a motion detection unit within a deep learning framework that iteratively refines the noisy labels during the training phase. Section VI delineates the design of the detection unit and the proposed framework.

IV. Overview of the Proposed Method

An overview of the proposed method for gesture recognition using sparse multichannel sEMG is shown in Fig. 2. It contains two parts. In the first part, we convert a windowed sEMG into an sEMG-Map which fuses the raw sEMG signals and their localized characteristic in the time-frequency domain. The map assembles the Locally Fused Units (LFUs) obtained from each sEMG channel vertically with the width the same as the window length (time). Each LFU maps the sEMG activities and the hierarchical multiresolution representation of the information contained in the sEMG signal in different detail levels by means of orthogonal wavelet, thus conveying a comprehensive piece of information about the electrical behavior of the recorded MUAPs. Furthermore, the assembled LFUs map the global spatiotemporal distribution among muscles that have been recorded through different channels (muscle's synergies). Then rely on the deep learning capacity to discover the elaborate structures and exploit the composite hierarchies in the fused sEMG-Map in order to find patterns that are repeatable among the same gesture yet discriminative across distinct gestures.

We constructed our fused sEMG-Map based on the following basic assumptions:

1) Sparse multichannel sEMG data can be thought of as poor-resolution spectral information.
2) Wavelets are effective measures for highlighting localized nonstationary signal features that are intrinsically prone to dynamic change in time-frequency contents.
3) Redundancy and multiple representation learning are core elements that are inherited in the design of state-of-the-art of deep learning, unlike the conventional machine learning methods that often use" shallow" architectures where redundancy is believed to obtain fault tolerance [4], [12].
4) EMG signal has a hierarchy's nature, where the firing of each individual fiber that composes the activated MU form MUAP, MUAPs gather into muscle's synergies, and muscle's synergies contribute to forming a particular movement.
5) Deep learning has an extraordinary capacity to discover elaborate structures in dimensionality data and exploit the composite hierarchies in natural signals where lower-level features comprise high-level ones.

In the second part, to deal with the label noise that mainly emerges from the erroneous rest-movement-rest sequence label, we proposed a novel data-centric method based on a CNN model (denoted as ALR-CNN) that aims to refine the mislabeling samples by correcting the rest-movement-rest sequence during the training phase. In brief, after each training epoch, a designated detection unit is used to estimate the samples where muscle activation intervals occurred based on a set of successive CNN's max-activated outputs. Then the rest-movement-rest sequence is refined according to the estimated intervals and fed back to the CNN model for further optimization in the next training epoch.

The design of the proposed ALR-CNN method follows two basic assumptions:

1) During training, the deep learning model becomes increasingly accurate.

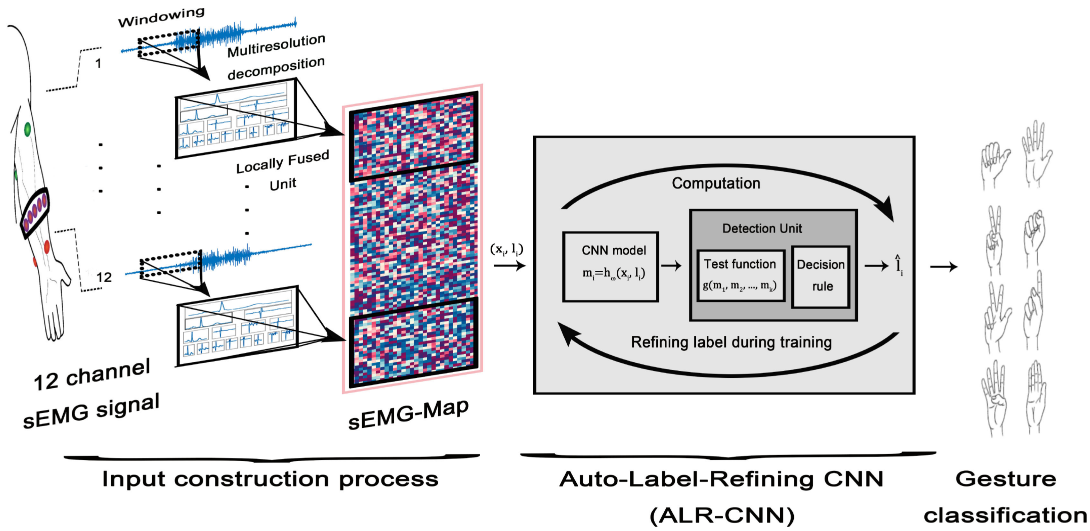

Fig. 2. Schematic of the proposed method for sEMG-based gesture recognition in the presence of label noise.

2) A set of successive learned max-activated outputs of a deep learning model that is robust against the noise and obtained high accuracy over a clean dataset is a reliable estimator for the occurrence of the change in the muscle activation pattern.

V. sEMG Image Fusion Through Multiresolution Decomposition

Deep learning methods have the capacity to process natural signals in their raw form. However, in the case of gesture recognition based on an image-classification framework, their performance on sparse multichannel sEMG is still far from optimal. That could be due to the limited information presented on the formed input image by sparse multichannel sEMG compared to the HD-sEMG that uses hundreds of electrodes. Sparse multichannel sEMG can be seen as poor-resolution multispectral data. Therefore, we proposed a data fusion method that attempts to enhance their spectral information by merging a coarse multiresolution decomposition by means of orthogonal wavelets through the recorded sEMG signals and forming a rich-detailed map image. The wavelet decomposition provides a hierarchical multiresolution representation of the information contained in the sEMG signal at different detail levels. Moreover, the wavelets are powerful tools to highlight localized characteristics of the nonstationary signal (e.g., EMG signal) that are inherently subject to dynamic change in time-frequency contents [37]. The proposed data fusion algorithm is detailed as follows:

A sliding window of 256 ms size is applied to the raw sEMG signal, where a multilevel wavelet decomposition technic is applied to the signal in each processing window. In practice, a four-level discrete wavelet transform (DWT) is applied by using the multiresolution analysis implementation introduced by Mallat [38]. Each wavelet packet coefficient is given by

$$W_{j,n,k} = < f, W^n_{j,k} > = < f, 2^{j/2} W^n(2^j t - k) > \quad (2)$$

where j is a scaling parameter, k is a translation parameter, and n is an oscillation parameter. For each channel, the raw sEMG signal and the multilevel decomposition form a Locally Fused Unit (LFU) that provides time-frequency localization and multiscale resolution. Thereby, the LFU presents a piece of comprehensive information about the electrical behavior of each recorded MUAP.

After that, all LFUs are gathered to form a map that provides a global representation of the spatiotemporal distribution of the recorded sEMG, thus conveying information about the muscle synergies that contribute to the particular movement of interest. The resolution of the map is denoted by $H \times W$, where H is the product of the number of electrodes and levels of decomposition (including the sEMG activities), and W is the length of the last level of WPT in the current window. Here, 12-electrodes $\times$ 5-levels of decomposition returns $H = 60$, and a window size of 256 ms returns $W = 544$, since f_s is $2k\ Hz$. Then, the map is linearly transformed into a grayscale image by converting the signal values to gray intensity, specifically, from [-0.25 mV, 0.25 mV] to [0, 1]. Then we created an RGB image by replicating the grayscale image three times.

We shift the window at an interval of 32 ms to obtain better real-time performance and serve as an augmentation to increase the number of images for the model training process. This process is consistently repeated for the whole signal length. Consequently, recognition of gestures based on fusion images turns into an image classification task that can be dealt with through supervised deep learning methods. Given an image, the classifier can predict the probability of each class after being trained with the labeled images.

VI. Auto-Label-Refining Convolutional Neural Network

The primary reason for label noise in gesture recognition based on sEMG arises from the erroneous rest-movement-rest sequence labeling. Motivated by the high accuracy and robustness of the CNN model and its generalization capabilities, we proposed an ALR-CNN to deal with the noisy label by refining the movement boundaries in a unified end-to-end learning framework. The pipeline of the ALR-CNN training framework for learning with noisy labels is illustrated in Fig. 2. The dataset was divided into folds according to the number of movement repetitions. Each fold was iteratively used to test the CNN model and self-refining the label sequence, while the remaining ones were used for the training phase. After each training epoch, the given labels and the prediction outputs of the model are used to refine the movement boundaries of the test dataset through an iterative label update strategy. The description of the iterative label update strategy is as follows.

Supposing the given rest-movement-rest label sequence vector of the current test dataset is composed of R movements separated by rest gestures. After each training epoch, the given labels sequence vector is divided into R overlapping investigating windows, where each window has the size M_r and contains three gestures (i.e., the r-th movement, the previous, and the following rest.). A test function g, which indicates possible events, is calculated from the current epoch CNN's output in the subsequent processing stage. Precisely, at each particular instant k, the test function g_k is calculated from the samples $m_k, m_{k+1}, \ldots, m_{k+L}$ within a sliding test window of size L that shifted sample by sample along the data sequence within the current investigating window. The test function g_k is calculated by counting the number of occurrences where CNN's output is predicted as the r-th movement in the k-th testing window. After all g functions in the current investigating window are calculated, the last k instant where the g function's value satisfies the condition of being a maximal is selected as the estimated onset t_{on}^r. The following decision rules summarize the selection strategy of the estimated onsets, and Algorithm 1 details the steps:

$$t_{on}^r = max\{1 \leq k \leq M_r : g_k \in D_r\}. \tag{3}$$

$$g_k = \sum_{i=k}^{k+L} [m_i = r] \tag{4}$$

$$D_r = \underset{1 \leq k \leq M_r}{\text{arg max}}\, g_k \tag{5}$$

The rationale for this rule is based on the observed confidence of the deep model that reasonably learned the muscle activation pattern of a clean dataset yet robust against the noisy label. However, using the max-activated output as the estimated label directly is risky due to the fact that hard samples can be taken as label noise. Therefore, using a set of successive learned max-activated outputs as an estimator for the occurrence of the change in the muscle activation pattern would reduce the risk of hard samples being taken as noisy labels. Note that selecting the last k instant where the g function's value satisfies the maximal condition helps prevent identifying a short burst as a movement. In contrast, choosing the proper size of the testing window L helps prevent identifying part of the actual movement as a short burst since the deep model may fail to predict the actual class for the entire movement length continually.

Algorithm 1: The strategy of selecting the onsets

Input 1 $Label$ ▷The current rest-movement-rest labels sequence, contained of n components and R movements
Input2 $predict$ ▷The predicted classes of the current epoch
Output t_{on} ▷The estimated onsets
Determine c_{on} ▷the indices of the current rest-movement-rest labels sequence onsets
Determine c_{off} ▷the indices of the current rest-movement-rest labels sequence offsets
for $r \leftarrow 1, R$ **do**
  Set $t_{on}(r) = 0$
  if $r = 1$ **then**
    Set $M_r(r) = (1 : c_{on}(r+1))$
  else if $r = l$ **then**
    Set $M_r(r) = (c_{off}(r-1) : n)$
  else
    Set $M_r(r) = (c_{off}(r-1) : c_{on}(r+1)$
  else if
  for $k \leftarrow 1, sizeM_r(r)$ **do**
    Calculate g_k ▷according to Eq. (4)
  end for
  Find $D_r set$ ▷according to Eq. (5)
  Estimate $t_{on}(r)$ ▷according to Eq. (3)
end for

After all t_{on}^r in all windows are estimated, the testing window starts to move backward where the testing function g_k^* is computed from the samples $m_{k-L}, m_{k-L+1}, \ldots, m_k$. The decision rule became as follows:

$$t_{off}^r = \min\{1 \leq k \leq M_r : g_k^* \in D_r^*\} \tag{6}$$

$$g_k^* = \sum_{i=k-L}^{k} [m_i = r] \tag{7}$$

$$D_r^* = \underset{1 \leq k \leq M_r}{\text{arg max}}\, g_k^* \tag{8}$$

Finally, the estimated boundaries are used to refine the rest-movement-rest label sequence of the current test dataset. After looping over folds, the samples with the new estimated labels are fed back to the CNN for further optimization of the model in the next training epoch. We started the refining process early (i.e., after the first epoch) based on the assumption that deep models learn the prominent patterns shared among the data samples before fitting the noisy labels [39], [40].

In this work, we set the sliding test window size L to be equivalent to the optimal movement length according to the experiment setup (i.e., 5 s). However, the flexibility of the estimated movement length is ensured by using two testing functions, g and g*. Moreover, the shortcoming of mislabeling an informative complex sample is limited since the estimated labels can be changed only between the rest class and the class of the r-th movement under consideration in the current investigating window and vice versa. However, for some gestures,

TABLE I

41 HAND GESTURES AND MOVEMENTS LABELED IN THREE NINAPRO DATASETS. FIRST COLUMN: 9 ISOMETRIC AND ISOTONIC HAND CONFIGURATIONS (I.E. G00:G08); SECOND COLUMN: 9 BASIC MOVEMENTS OF THE WRIST (I.E. G9:G17); THIRD AND FOURTH COLUMNS: 23 GRASPING AND FUNCTIONAL MOVEMENTS (I.E. G18:G40)

G00-Rest; G01-Thumb up; G02-Extension of index and middle,flexion of the others; G03-Flexion of ring and little finger, extension of the others; G04-Thumb opposing base of little finger; G05-Abduction of all fingers; G06-Fingers flexed together in fist; G07-Pointing index; G08-Adduction of extended fingers; G09-Wrist supination (axis: middle finger); G10-Wrist pronation (axis: middle finger); G11-Wrist supination (axis: little finger); G12-Wrist pronation (axis: little finger); G13-Wrist flexion; G14-Wrist extension; G15-Wrist radial deviation; G16-Wrist ulnar deviation;G17-Wrist extension with closed hand; G18-Large diameter grasp; G19-Small diameter grasp (power grip); G20-Fixed hook grasp; G21-Index finger extension grasp; G22-Medium wrap; G23-Ring grasp; G24-Prismatic four fingers grasp; G25-Stick grasp; G26-Writing tripod grasp; G27-Power sphere grasp; G28-Three finger sphere gras; G29-Precision sphere grasp; G30-Tripod grasp; G31-Prismatic pinch grasp; G32-Tip pinch grasp; G33-Quadpod grasp; G34-Lateral grasp; G35-Parallel extension grasp; G36-Extension type grasp; G37 Power disk grasp; G38-Open a bottle with a tripod grasp; G39-Turn a screw; G40-Cut something.

the estimated motion may start slightly after that actual motion, end early before the actual motion, or both, probably due to the common reach-to-gesture phases between some gestures. For example, the reaching phase of the "Small diameter grasp (power grip)" gesture (denoted as G19 in Table I) could be identically part of the reaching phase of the "Fixed hook grasp" gesture (denoted as G20 in Table I). That is also true for the releasing phase.

VII. EXPERIMENTS AND RESULTS

In our work, we designed a series of experiments using three large-scale public databases in order to (i) evaluate the effectiveness of training a deep model on the constructed sEMG-Map for the gesture recognition task; (ii) assess the quality of the refined label by our proposed ALR-CNN method; (iii) evaluate the effectiveness of the proposed gesture recognition framework for learning with label noise; and (iv) conduct a performance comparison of the proposed framework with the state-of-the-art methods.

A. Dataset

The data used in this study consists of three Ninapro data sets, i.e., DB2, DB3, and DB7, including sEMG records of 73 participants (13 trans-radial amputees, 60 intact people) performing 41 gestures with six repetitions for each. The collection of movements was chosen from hand taxonomy, robotics, and rehabilitation literature [41]–[46] in order to include the most common gestures observed in activities of daily living (ADL). Table I addresses the different types of movements. The Ninapro DB2 contains data collection of 40 healthy participants, the DB3 data acquisition contains 11 trans-radial amputees, and the DB7 data collection comprises 20 healthy participants and two trans-radial amputees. All data sets were collected with 12-electrodes channels (Delsys Trigno wireless system). The sampling frequency is 2000 Hz. For more information about the data acquisition and protocol, please refer to [10] and [47].

B. Experimental Setup

For all experiments, a deep residual neural network (ResNet) [48], especially, ResNet-18, was chosen as the CNN model in Fig. 2 for the image classification task. The ResNet-18 model pretrained on ImageNet [49] was used to learn the gesture classification task. The last two layers (i.e., the last learnable and final classification layers) have been replaced with new layers adapted to the task of 41 gestures classification. The model takes an sEMG-Map as input and then outputs a gesture label together with the probabilities for other gestures. The cross-entropy loss was used as the loss function for multi-class classification. Random searching and manual tuning [50] was used to identify hyper-parameters in the small data sets of randomly chosen subjects. The stochastic gradient descent was selected as the optimizer with a momentum equal to 0.9. The other hyperparameters were set as follows: a batch size of 32, an epoch number of 15, a weight decay of 0.0001, and the first fifteen of thirty logarithmically spaced points between decades 0.01 and 0.001 set for the learning rate.

For all experiments, we evaluated the intra-subject gesture recognition performance. We excluded the first trials from the evaluation metrics, which we considered a pre-training stage; to increase the subject's familiarity with the conducted exercises, reduce the computational cost and accelerate our experiments. For the remaining five trials, we conducted Leave-One-Trial-Out Cross-Validation (LOTOCV): the data from each subject's trial were used in turn as the test data, and the CNN was trained using the data of the remaining trials.

The classifier's performance in terms of Classification Accuracy (CA) is defined as:

$$\text{CA} = \frac{\text{Number of correct classification}}{\text{Total number of test samples}} \times 100\%, \tag{9}$$

The gesture recognition accuracy was averaged over all trials and subjects.

For the experiment in VII-C, the robustness of the classifier was measured using the Relative Loss of Accuracy (RLA), which is defined as:

$$RLA_{HNL} = \frac{A_{LNL} - A_{HNL}}{A_{LNL}}, \tag{10}$$

where A_{LNL} is the accuracy of the classifier with a low noise level, and A_{HNL} is the accuracy of the classifier with a high noise level.

For the experiment in VII-D, we used Detection Accuracy (DA), degree of over detection (OD), degree of under detection (UD), and the F1 score to evaluate the quality of the refined labels. These metrics were given as percentages and were defined in terms of the following values:

1) True positive (TP): sEMG signal samples are categorized as a movement by both the visual inspection and the algorithm.
2) False positive (FP): sEMG signal samples are categorized as a movement by the algorithm and not by the visual inspection.

TABLE II
PERFORMANCE COMPARISON OF THE CNN MODEL TRAINED ON SEMG-MAPS AT DIFFERENT LABEL NOISE LEVELS (I.E., THE HIGH NOISE LEVEL CONTAINED ON GLR LABEL SEQUENCE AND THE LOW NOISE LEVEL OF THE CROPPED LABEL SEQUENCE), AND ITS RLA AT THE GLR NOISE LEVEL

Database	$A_{cropped}$	$A_{uncropped}$	RLA
first five subjects of DB2	89.03	88.94	0.001
first five subjects of DB7	92.6	92.6	0

3) True negative (TN): sEMG signal samples are not categorized as a movement by both the visual inspection and the algorithm.
4) False negative (FNthe) : sEMG signal samples are not categorized as a movement by the algorithm and categorized as a movement by the visual inspection.

The amount of agreement between the algorithm and the visualized labels in categorizing the sEMG signal as movement or rest correctly was evaluated as DA and defined as follows:

$$DA = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{11}$$

F_1 score was regarded as a combined assessment of the algorithm's sensitivity and precision and defined as follows:

$$F_1 = \frac{2 \times TP}{2 \times TP + FN + FP} \times 100\%, \tag{12}$$

OD refers to the degree to which the algorithm recognized the onset early and detected the offset late with respect to the visualized labels and defined as follows:

$$OD = \frac{FP}{TP + FN} \tag{13}$$

UD was defined as the degree to which the algorithm recognized the onset late and detected the offset early with respect to the visualized labels and defined as follow:

$$UD = \frac{FN}{TN + FP} \tag{14}$$

C. Performance of the Model Trained With sEMG-Maps

This subsection introduces the evaluation conducted to assess the performance of the CNN model that was trained on the constructed sEMG-Maps in terms of accuracy and robustness.

In this experiment, first, we balanced the rest class according to the number of movement repetitions. Then, in order to investigate the effects of the samples that have a potential high label noise on the performance of the gesture recognition framework, we set each movement (class) to last for 2 s and discarded the transition processes samples between classes where label noise is expected to be high. Fig. 3 compares the gesture recognition accuracy achieved on both the cropped (i.e., contains less label noise) and uncropped movement labeled by the GLR algorithm (i.e., contains more substantial label noise) for each dataset.

Furthermore, to measure the model's sensitivity to data corruption (that caused by the impact of noisy labels), we trained the model on both; the uncropped and cropped movements and tested it on the cropped movements for a sub-dataset that included the first five subjects of DB2 and DB7 databases. Table II shows the achieved accuracy for both the cropped

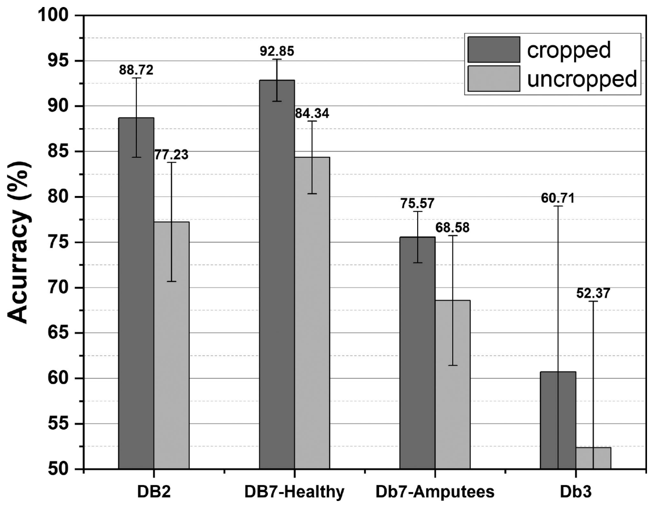

Fig. 3. Gesture recognition results achieved by the sEMG-Map for both cropped and uncropped datasets.

TABLE III
AGREEMENT AND DESCRIPTIVE STATISTICS FOR THE DETECTION METHODS COMPARED WITH VISUAL

Measure	Average ± Std %		
	Original	GLR	ALR
DA	65.33 ± 0.95	71.18 ± 0.89	**90.00 ± 2.52**
F1 Score	61.67 ± 1.19	65.54 ± 0.70	**81.07 ± 4.07**
UD	**0.54 ± 0.72**	1.21 ± 0.59	9.71 ± 2.15
OD	121.56 ± 8.44	99.00 ± 3.87	**10.78 ± 3.91**

and uncropped movements and the RLA associated with the uncropped movements.

D. Evaluation of the Proposed Auto-Label-Refining CNN

In this subsection, we attempt to validate the effectuality of the proposed ALR-CNN method on cleaning the label noise by realigning the movement boundaries. However, one hindrance of real data is that the true response of the movement's onset is unknown. Therefore, we assessed our method on a relative basis. Specifically, we visually corrected the movement's boundaries for a sub-dataset to estimate a reference line; then, we conducted a comparison between the proposed ALR-CNN method and the two labels sequences included in the NinaPro databases' files (i.e., the original labels and the refined labels by the GLR algorithm.) with respect to the visually estimated labels. The sEMG signals of the first amputee subject from the DB3 database were used in this subsection for the validation. In total, the boundaries of 145 movements were visually corrected and manually allocated onset and offsets of each movement.

Fig. 4 illustrates part of the raw sEMG signals and the corresponding events of each labeling method. Moreover, the statistical results in terms of an average and standard deviation of DA, F_1 score, UD, and OD of the proposed ALR-CNN method, GLR method, and the original labels sequence against the visually inspected labels are shown in Table III.

In general, imbalanced data indicates the realistic performance of the gesture recognition method where all data samples are included according to the realistic setting of the conducted exercises but suffers from the bias effect. On the other hand, balanced data gives a more accurate interpretation of the true performance in the case of clean datasets. However, for datasets that contain noisy labels, the presence of UD error rate shall lead to overestimating the true performance, and the presence of OD error rate shall lead to degenerating the true performance. Therefore, to better interpret the performance of our method, we calculated the classification accuracy for both balanced and imbalanced datasets. The results are shown in Fig. 5. For more details of the model performance during the self-correct training procedure, Fig. 6 plots the training and testing curves for both the balanced and imbalanced data of subject one in DB3. Moreover, we compared the classification accuracy obtained by discarding transition samples using the GLR method and the proposed ALR-CNN method over each dataset, see Table IV.

TABLE IV
GESTURE RECOGNITION ACCURACIES ACHIEVED ON NINAPRO DATABASES USING DIFFERENT CLEANING LABEL NOISE METHODS

Database	Cleaning method		
	GLR	$ALR-CNN$	$Cropping$
DB2	77.23%	87.85%	88.72%
DB7-Healthy	84.34%	90.00%	92.85%
DB7-Amputee	68.58%	73.12%	75.57%
DB3	52.37%	55.98%	60.71%

E. Computation Cost

The proposed method was implemented in Matlab (Mathworks Inc., USA) with the toolbox of deep learning and executed in parallel mode using Parallel Toolbox on a workstation with dual Intel(R) Xeon(R) Gold 5118 CPU (2.30 GHz), 64 G RAM, and one TITAN RTX GPU. A 512-sample analysis window corresponds to a processing delay of roughly 110 ms (i.e., about 80 ms for generating the sEMG-Map and 25 ms for classification). Moreover, about 75 minutes were needed to train the model using the GPU.

F. Comparison With the State-of-the-Arts

To demonstrate the benefits of integrating the sEMG-Map and ALR inside a deep learning framework, a comparison with the state-of-the-art gesture recognition using sEMG was performed. Table V summarizes the experimental results from earlier works as well as those obtained using our method on three sparse multichannel sEMG datasets: NinaPro DB2, NinaPro DB3, and NinaPro DB7. Moreover, Fig. 7 shows the accuracy of this work using different relabeling methods on each hand amputated subject in DB3 and DB7 and also provides a comparison between our work and that of Zhai *et al.*.

VIII. DISCUSSION

In this paper, we presented a novel gesture recognition framework to improve the performance of the MCI system. The framework consists of two parts. In the first part, we improved the input-output function flexibility by constructing a novel

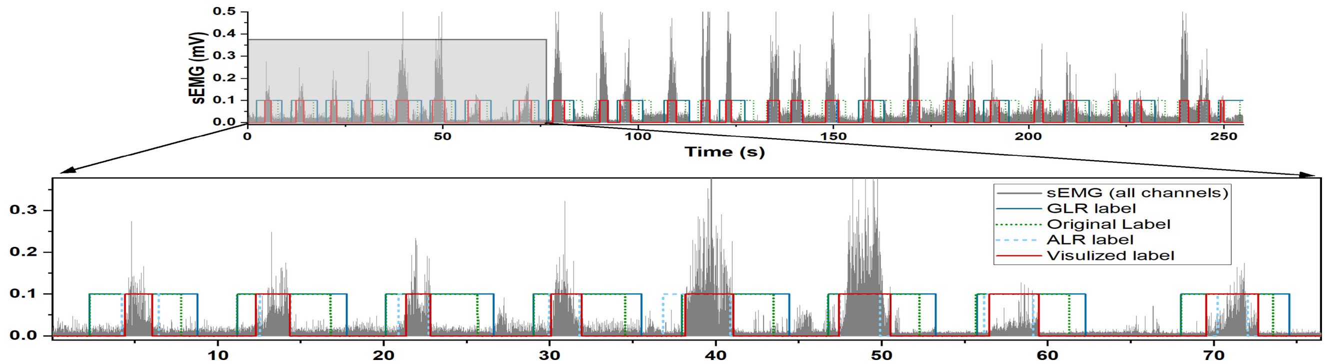

Fig. 4. Part of the raw sEMG signals obtained in a reaction time experiment.

TABLE V
GESTURE RECOGNITION ACCURACIES ACHIEVED BY THE PROPOSED METHOD COMPARED WITH THOSE OF THE EXISTING WORKS ON 3 sEMG DATABASES

Work	Input	Classifier	Relabeling method	Database	# of movements to be classified	Rest gesture	Accuracy
Atzori *et al.* [10]	classical features	Random forest	GLR	NinaPro DB2	50	Balanced	75.3%
Atzori *et al.* [19]	raw-sEMG Image	CNN	GLR	NinaPro DB2	50	Balanced	60.3%
Zhai *et al.* [21]	sEMG-spectrograms Image	CNN	GLR	NinaPro DB2	50	Balanced	78.7%
Zhai *et al.* [21]	sEMG-spectrograms Image	CNN	GLR	NinaPro DB2	41	Balanced	76.34%
Wei *et al.* [22]	Multi-View	Multi-CNN	GLR	NinaPro DB2	50	unknown	83.7%
this work	sEMG-Map	CNN	GLR	NinaPro DB2	41	Balanced	77.23%
this work	sEMG-Map	CNN	ALR-CNN	NinaPro DB2	41	Balanced	**87.9%**
Atzori *et al.* [10]	classical features	Random forest	GLR	NinaPro DB3	50	Balanced	46.3%
Atzori *et al.* [19]	raw-sEMG Image	CNN	GLR	NinaPro DB3	50	Balanced	38.09%
Zhai *et al.* [21]	sEMG-spectrograms Image	CNN	GLR	NinaPro DB3	50	Balanced	> 54.06%
Wei *et al.* [22]	Multi-View	Multi-CNN	GLR	NinaPro DB3	50	unknown	**64.3%**
this work	sEMG-Map	CNN	GLR	NinaPro DB3	41	Balanced	52.37%
this work	sEMG-Map	CNN	ALR-CNN	NinaPro DB3	41	Balanced	56%
Krasoulis *et al.* [47]	classical features	LDA	GLR	NinaPro DB7	41	Balanced	58.3%
Wei *et al.* [22]	Multi-View	Multi-CNN	GLR	NinaPro DB7	41	unknown	88.3%
this work	sEMG-Map	CNN	GLR	NinaPro DB7	41	Balanced	82.91%
this work	sEMG-Map	CNN	ALR-CNN	NinaPro DB7	41	Balanced	**88.5%**

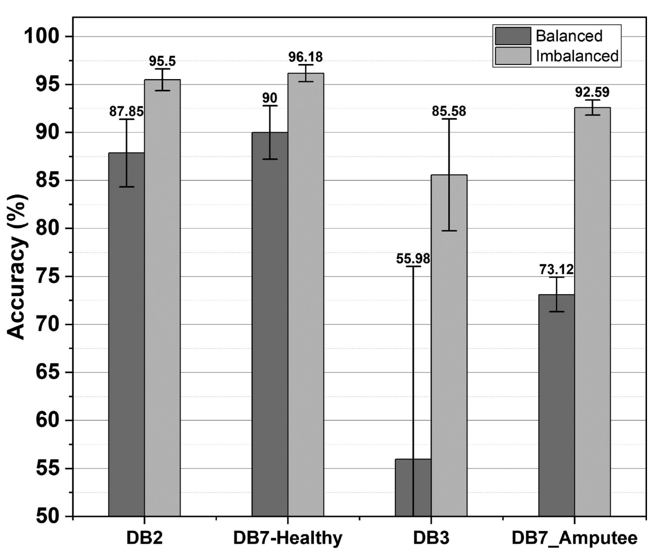

Fig. 5. Gesture recognition results achieved by the proposed method for both imbalanced an balanced datasets.

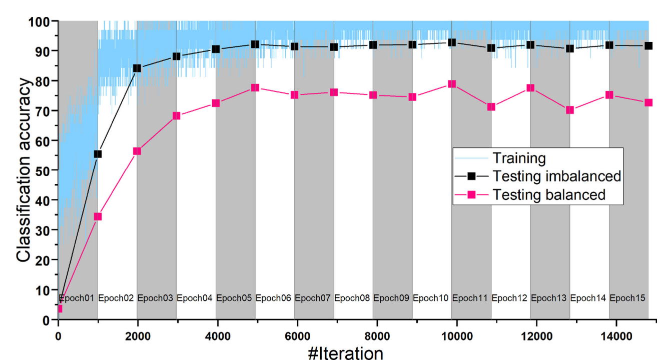

Fig. 6. Analysis of the ALR-CNN on amputee subject 1 under DB3. the model trained on four-folds and tested on the reminded one for both balanced and imbalanced data.

image-map input (i.e., sEMG-Map) and fed it into a CNN model. The sEMG-Map was formed in a way that:

1) enriches the spectral information of the poor-resolution sparse multichannel sEMG signals.
2) follows the hierarchy's nature of the EMG signals in terms of their contribution to a particle movement (gesture) of interest.
3) facilitates the implantation of CNN models to benefit from their capabilities of exploiting the composite hierarchies in natural signals.

In the second part, we dealt with the label noise by proposing a novel data-centric method (i.e., ALR-CNN) that:

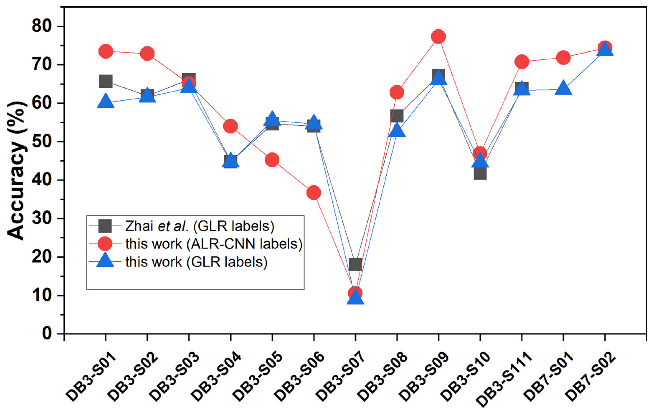

Fig. 7. Comparison subject-to-subject accuracy in amputee subjects.

1) detects the motion boundaries based on the learned features of the muscle activation pattern in terms of a successive set of max-activated outputs.
2) refines the mislabeled samples in a unified end-to-end learning framework.

Regarding the performance of the CNN model that was trained on the constructed sEMG-Maps, statistical results presented in Fig. 3 show that the model achieved an average accuracy of 77.23% on DB2 (uncropped) with 16.93% improvement over the baseline gesture recognition accuracy obtained by [19] using an end-to-end deep learning framework and 20.03% improvement over the recognition accuracy achieved by [22] using a single-stream CNN that fed by WPT coefficients. Moreover, the classification accuracy significantly improved over the clean datasets (cropped), which validates the effectiveness of the proposed method sEMG-map generation. In addition, the experimental results establish reasonable confidence in the model generalization capability and robustness against noisy labels (see Table II). Therefore, that motivated us to design the ALR-CNN method to iteratively refine the movement's boundaries during the training processes instead of discarding the transition processes between classes, thus obtaining a more realistic performance.

In respect of the quality of the refined label sequences, the statistical results presented in Table III show that our proposed ALR-CNN method has a higher degree of agreement with the visualized labels. Up to 90% of the signal's samples were correctly distinguished into movement and rest classes (about 25% and 19% increase over the original labels sequence and the GLR method, respectively). ALR-CNN also outperforms the other two methods in terms of the harmonic mean of the algorithm's sensitivity and precision (i.e., F_1 score) with an improvement of about 20% and 15% compared to the original labels sequence and the GLR method, respectively. Nonetheless, GLR and the original labels sequence show a very low UD error rate; however, this comes with an excessive increase in OD error rate, 99.00%, and 121.56%, respectively. In comparison, the results suggest that the proposed method has a degree of tolerance to the tradeoff between the UD and OD error rates. Our proposed ALR-CNN method shows about 10% error rate for both UD and OD.

Moreover, Table III shows that there is about 30% of label noise exists in the sub-dataset using the GLR method. However, in such scenarios, if the data is balanced against the rest class, the rest class bias will be eliminated; on the other hand, the label noise will be increased to about 50%; since the OD rate is 99% which shall seriously degenerate the performance of gesture recognition methods. In comparison over the same sub-dataset, the proposed ALR-CNN method minimizes the noise labels to about 10% for both imbalanced and balanced data.

The classification results described in the literature are often imbalanced, and more often, it is not mentioned if the results are balanced. However, calculating both balanced and imbalanced accuracies better interprets the model's true performance and the quality of balanced datasets. The statistical results presented in Fig. 5 show that the gap between balanced and imbalanced datasets is reasonable due to the expected ratio between the number of samples labeled as rest gesture to the total number of other gestures samples.

In addition, Fig. 6 presents the model performance during a fold self-correct training procedure for the amputee subject 1 under DB3. The obtained curves indicate that the model started by learning the dominated features and obtained a reasonable result after a few epochs, then began to slightly fluctuate due to the change of the estimated movement borders after each epoch.

Furthermore, we validate the effectiveness of the proposed ALR-CNN in improving the model performance by comparing the classification accuracies obtained by discarding transition samples, the GLR method, and the proposed ALR-CNN method over each dataset. Table IV shows that the obtained results using the ALR-CNN method are slightly lower than those obtained over the cropped datasets. This is expected since the initial and the final parts of movements are more difficult to classify than the steady parts where the gesture is maintained. However, the classification accuracy is significantly improved compared to the GLR method, which validates the effectiveness of our proposed method in handling the label noise and improving the performance of the gesture recognition task.

Finally, we demonstrated the effectuality of our proposed framework by conducting a comparison with the state-of-the-art gesture recognition using sEMG. It is worth noting that major differences between studies, such as adopting different data acquisitions and protocols, number of movements, type of movements, and evaluation procedure, could restrict a fair comparison. However, most previous works under DB2 and DB3 dealt with 50 movements, including the 41 movements conducted in this work (i.e., 17 basic movements, 23 grasping and functional movements, and rest class) in addition to nine force patterns. In general, the classifier performance is expected to degrade as the number of movements increases. However, force patterns are relatively easy classes compared to grasping and functional movements or even basic movements. In his work, Zhai *et al.* [21] showed that including the nine force patterns classes increased the overall accuracy by 2.36% over NinaPro DB2. Moreover, Table V shows that our proposed method outperforms the state-of-the-art methods except for the work of Wei *et al.* under NinaPro DB3, and we obtained an average classification accuracy of (56.00±20.08)%, which is inferior to the average classification accuracy described by Wei *et al.*

(64.1± 6.7)%. Several factors can determine the classification accuracy difference. At the outset, it is not clear whether the rest class was balanced in Wei *et al.* 's work. Moreover, the performance of hand amputated subjects should interpret with caution due to the differences in their clinical parameters, which can strongly affect their capabilities to reproduce movements. However, only Zhai *et al.* reported a subject-to-subject performance comparison for the hand amputated subjects under the DB3 database in his work. Fig. 6 shows the performance of each hand amputated subject in DB3 and DB7 and compares our findings with Zhai *et al.* 's work. As can be seen, the proposed ALR-CNN method improved the performance on most subjects (e.g., six subjects exceeded 70%, and eight subjects obtained an average accuracy of 71.08%).

Nevertheless, the ALR-CNN method failed to improve the classification accuracy for a few subjects. The proposed ALR-CNN method depends upon the CNN's prediction capabilities to refine the movements' borders. Therefore, for some subjects with a very low accuracy, the ALR-CNN method suffers to correctly estimate the movements' borders which may lead to further degeneration in the performance. For example, the ALR-CNN method and visual inspection fail to reveal the occurrence of most movements performed by the amputee subject 7, probably because the lost of entire forearm affects the ability to reproduce movements which can be recorded through the adopted acquisition setup.

The size of the processing window and the step increment was chosen to balance the tradeoff between the amount of time available for sEMG signal collection and analysis and the responsiveness of the prosthesis. However, for a real-time myoelectric system, the controller delay should be kept below 300 ms [51]. Therefore, we set the size of the processing window to be 256 and the increment step to be 32 ms. 256 ms window is long enough to maximize the classification accuracy and copes with any expecting buffer (up to eight windows) without discarding any portion of the data by selecting the last generated window for further processing. Since each analysis window is incremented at 32 ms, the time needed to produce a new decision equals 32 ms if the processing delay time (i.e., the time needed for generating the sEMG-Map and discrimination) is less or equal to the increment step. Otherwise, the time needed to produce a new decision equals the processing delay time. In this work, the time taken to make a decision was around 110 ms, which is below 300 ms for real-time control.

Moreover, the obtained results on the conducted datasets consisting of a relatively large number of participants and gestures indicate that the proposed framework could be translatable and generalizable to other datasets in the light of an intra-subject point of view.

A. Limitations and Future Work

Heavy training burden and implementation under user-specific conditions are two major limitations of our work. Therefore, future work will focus on building a user-independent gesture recognition framework by integrating the powerful generalization ability of deep learning networks and transfer learning methods in order to increase the generalizability and reduce the training burden.

IX. Conclusion

In this paper, we proposed a novel gesture recognition framework to improve the performance of an sEMG-based gesture recognition system. We introduced the concept of sEMG-Map to enhance the spectral information of the sparse multichannel sEMG. We also designated an event detection unit and merged it within a CNN model to self-correct the noisy labels during the training stage. Furthermore, we evaluated our proposed framework on three large-scale public databases, including 60 intact subjects and 13 hand amputated subjects for the recognition of 41 hand gestures. The statistical results demonstrate the effectuality of the proposed method and show high classification accuracy and robustness against label noise.

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